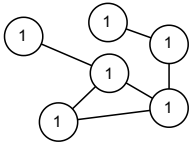
a

G

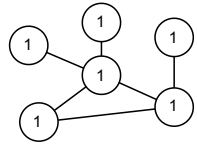


$$\beta = \left\{ c^{(k)}(v), \left\{ c^{(k)}(u) \right\}_{u \in N(v)} \right\}$$

initial color $c^{(0)}(v) = 1$ for every node

β	HASH(β)	Count
{1}	1	6

G′



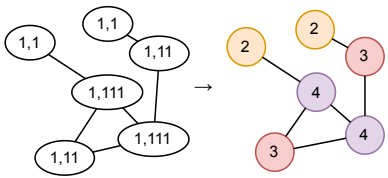
$$\beta = \left\{ c^{(k)}(v), \left\{ c^{(k)}(u) \right\}_{u \in N(v)} \right\}$$

initial color $c^{(0)}(v) = 1$ for every node

β	HASH(β)	Count
{1}	1	6

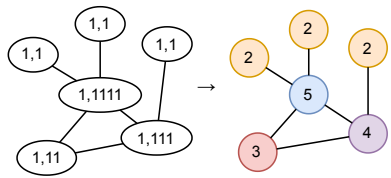
b

k = 0



β	HASH(β)	Count
{1,{1}}	2	2
{1,{1,1}}	3	2
{1,{1,1,1}}	4	2
{1,{1,1,1,1}}	5	0

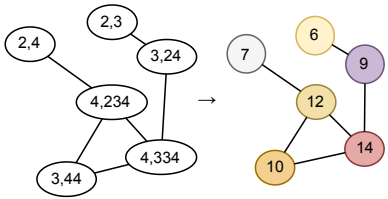
k = 0



β	HASH(β)	Count
{1,{1}}	2	3
{1,{1,1}}	3	1
{1,{1,1,1}}	4	1
{1,{1,1,1,1}}	5	1

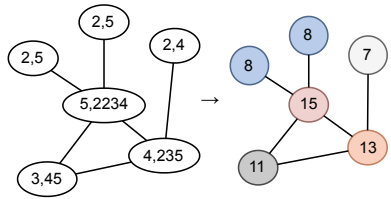
c

k = 1



β	HASH(β)	Count
{2,{3}}	6	1
{2,{4}}	7	1
{2,{5}}	8	0
{3,{2,4}}	9	1
{3,{4,4}}	10	1
{3,{4,5}}	11	0
{4,{2,3,4}}	12	1
{4,{2,3,5}}	13	0
{4,{3,3,4}}	14	1
{5,{2,2,3,4}}	15	0

k = 1



β	HASH(β)	Count
{2,{3}}	6	0
{2,{4}}	7	1
{2,{5}}	8	2
{3,{2,4}}	9	0
{3,{4,4}}	10	0
{3,{4,5}}	11	1
{4,{2,3,4}}	12	0
{4,{2,3,5}}	13	1
{4,{3,3,4}}	14	0
{5,{2,2,3,4}}	15	1

d

Count vector over the colors of rounds 0 and 1

$$\phi(G) = [6, 2, 2, 2, 0, 1, 1, 0, 1, 1, 0, 1, 0, 1, 0]$$

Count vector over the colors of rounds 0 and 1

$$\phi(G') = [6, 3, 1, 1, 1, 0, 1, 2, 0, 0, 1, 0, 1, 0, 1]$$

e

WL subtree kernel of each graph with itself

$$\phi(G)^\top \phi(G) = 54$$

WL subtree kernel of each graph with itself

$$\phi(G')^\top \phi(G') = 56$$

f

WL test: the count vectors differ after one round, so the graphs are not isomorphic

$$\phi(G)^\top \phi(G') = 47$$